

5. Pancreatitis is a disease that causes inflammation of the pancreas and can result in a very low rate of enzyme production and secretion into the small intestine. Enzymes can be given to patients as a treatment for this disease. The enzymes are taken as a slow-release capsule which is not affected by acid.

(a) State the function of an enzyme. [1]

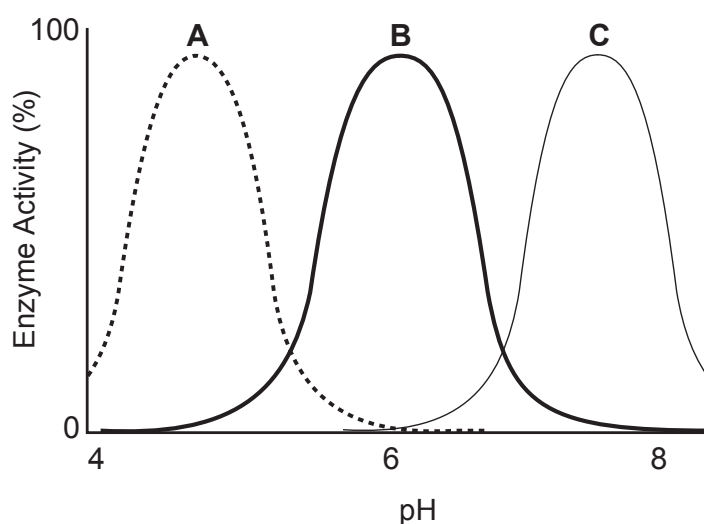
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(b) Suggest a reason why the enzyme is not given to the patient as a liquid. [2]

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(c) The activity of enzymes **A**, **B**, and **C** at different pH levels are shown below.



Explain whether enzymes **A**, **B**, or **C** are suitable for use in treating a patient with pancreatitis.

[4]

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(d) Enzymes are also used in the manufacture of biological washing powders. Biological washing powders contain proteases, carbohydrases and lipases.

- (i) Suggest why washing powders need these three types of enzymes to remove food stains from clothes. [1]

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- (ii) Explain why washing at 80 °C using biological washing powders would not be recommended for removing food stains. [4]

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